



6.1 Introduction to Combinational Circuits :

SPPU : April 10, Oct. 15

University Questions

Q. 1 Give one example of combinational circuit.

(April 10)

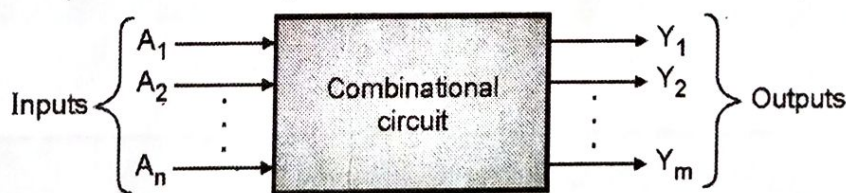
Q. 2 What is combinational circuit ? Define decoder.

(Oct. 15, 2 Marks)

- The digital circuits in general are classified into two categories namely :
 1. Combinational logic circuits
 2. Sequential logic circuits.

Definition :

- A combinational circuit is a logic circuit the output of which depends only on the combination of the inputs. The output does not depend on the past value of inputs or outputs. Hence combinational circuits do not require any memory (to store the past values of inputs or outputs).
- The combinational circuit is a digital system the output of which at any instant of time, depends only on the levels present at its input terminals.
- The combinational circuits do not use any memory. Hence the previous state of input does not have any effect on the present state of the circuit.
- The sequence in which the inputs are being applied also does not have any effect on the output of a combinational circuit.
- A combinational circuit is a logic circuit the output of which depends only on the combination of the inputs. The output does not depend on the past value of inputs or outputs. Hence combinational circuits do not require any memory (to store the past values of inputs or outputs).
- The block diagram of a combinational circuit is shown in Fig. 6.1.1.
- A combinational circuit can have a number of inputs and a number of outputs. The circuit of Fig. 6.1.1 has "n" inputs and "m" outputs.



(C-332) Fig. 6.1.1 : Block diagram of a combinational circuit

- A combinational circuit is made up of logic gates. These gates are connected suitably between the inputs and outputs of the combinational circuit.
- A combinational circuit operates in three steps :
 1. It accepts n-different inputs.
 2. The combination of gates operates on the inputs.
 3. "m" different outputs are produced as per requirement.

Examples :

- Following are some of the examples of combinational circuits :

1. Adders, subtractors
2. Comparator
3. Code converters
4. Encoders, decoders
5. Multiplexers, demultiplexers

244 Analysis of a Combinational Circuit .

Examples :

- Following are some of the examples of combinational circuits :

1. Adders, subtractors
2. Comparator
3. Code converters
4. Encoders, decoders
5. Multiplexers, demultiplexers

**6.2.1 Half Adder :**

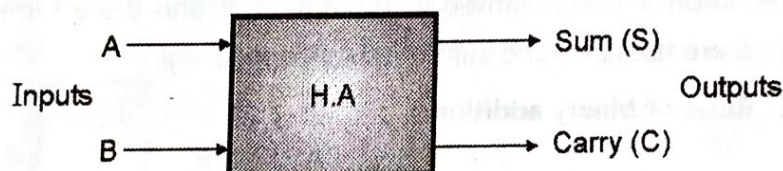
SPPU : Oct. 11, April 14, April 15, Oct. 15, April 16, Oct. 17, April 18

University Questions

- Q. 1 Give the working of half adder with neat logic diagram. Give the truth table of half adder. (Oct. 11, 4 Marks)
- Q. 2 Draw the truth table and logic diagram of half adder and half subtractor. (April 14, Oct. 15, April 18, 4 Marks)
- Q. 3 Write the truth table for half adder. (April 15, 2 Marks)
- Q. 4 Draw the circuit diagram of Half adder and Half subtractor and write truth tables of both. (April 16, 4 Marks)
- Q. 5 Draw logic diagram for half adder. (Oct. 17, 2 Marks)

Definition and Truth table :

- Half adder is a combinational logic circuit with two inputs and two outputs. It is the basic building block for addition of two "single" bit numbers. This circuit has two outputs namely "carry" and "sum". The block diagram of half adder is as shown in Fig. 6.2.2(a).
- The half adder circuit is supposed to add two single bit binary numbers A and B. Therefore the truth table of a half adder is as shown in Fig. 6.2.2(b).

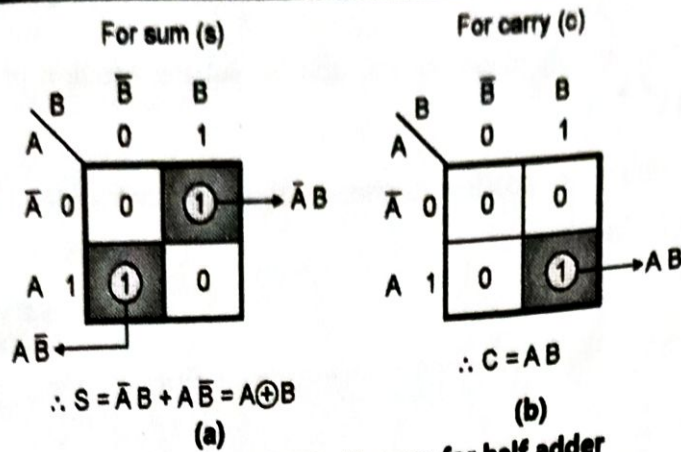


Inputs		Outputs	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

(C-344) Fig. 6.2.2 : Half adder

Design using k-maps and equations :

- K-maps for carry and sum outputs are as shown in Figs. 6.2.3(a) and (b).



(C-345) Fig. 6.2.3 : K-maps for half adder

Simplified expressions for the two outputs are :

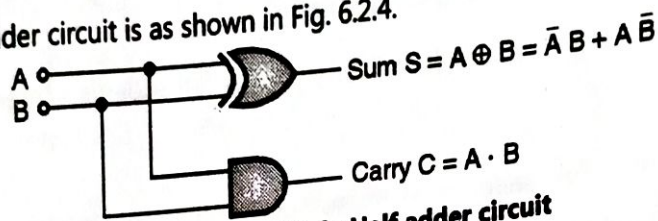
...(6.2.1)

$$\left. \begin{aligned} S &= \bar{A}B + A\bar{B} = A \oplus B \\ C &= AB \end{aligned} \right\}$$

(C-6374)

Logic Diagram :

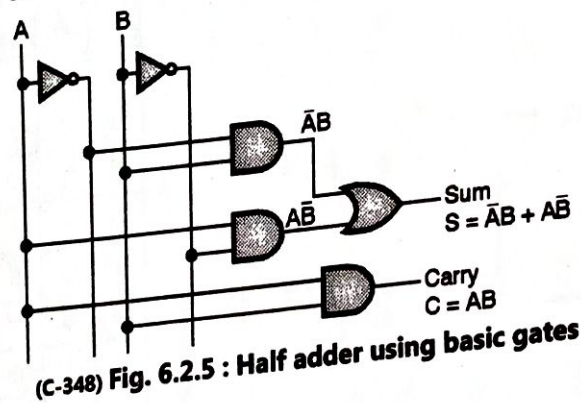
Hence the half adder circuit is as shown in Fig. 6.2.4.



(C-346) Fig. 6.2.4 : Half adder circuit

Half adder using Basic gates :

The half adder circuit can be designed using basic gates is as shown in Fig. 6.2.5.



(C-348) Fig. 6.2.5 : Half adder using basic gates

Disadvantage of Half Adder :

The principle of adding two 2-bit numbers A and B is as shown in Fig. 6.2.6.
Let number A = $A_1 A_0$ and Number B = $B_1 B_0$

Then the addition should take place as follows :

$$\begin{array}{r} A_1 \quad A_0 \leftarrow \text{Number A} \\ + \quad B_1 \quad B_0 \leftarrow \text{Number B} \\ + \quad \boxed{C_0} \leftarrow \text{Carry generated from the addition } (A_0 + B_0) \\ \hline \boxed{C_1} \quad S_1 \quad S_0 \end{array}$$

(C-347) Fig. 6.2.6 : Two bit binary addition



- A half adder can add A_0 and B_0 to produce S_0 and C_0 . But the addition of next bits requires the addition of A_1 , B_1 and C_0 .
- The addition of three bits is not possible to perform by using a half adder. Hence we cannot use a half adder in practice.

SPPU : April 14, Oct. 14

6.2.2 • Full Adder :

University Questions

Q. 1 Write the truth table of full adder.

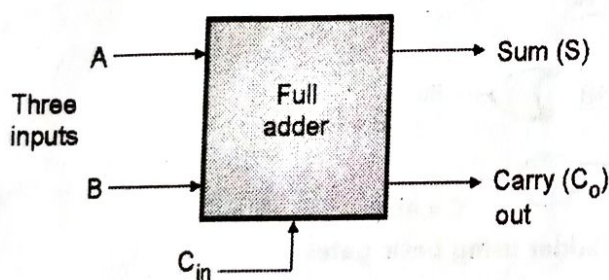
(April 14, 2 Marks)

Q. 2 Draw and explain the logic diagram of 3 bit full adder.

(Oct. 14, 4 Marks)

Definition and truth table :

- Full adder is a three input two output combinational logic circuit which can add three single bits applied at its input to produce Sum and Carry outputs
- It can add two one-bit numbers A and B, and carry C_{in} . The full adder is a three input and two output combinational circuit.
- The block diagram of a full adder is as shown in Fig. 6.2.7(a) and its truth table is given in Fig. 6.2.7(b). It has been designed to overcome the drawback of Half Adder circuit.



(a) Block diagram

Inputs			Outputs	
A	B	C_{in}	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

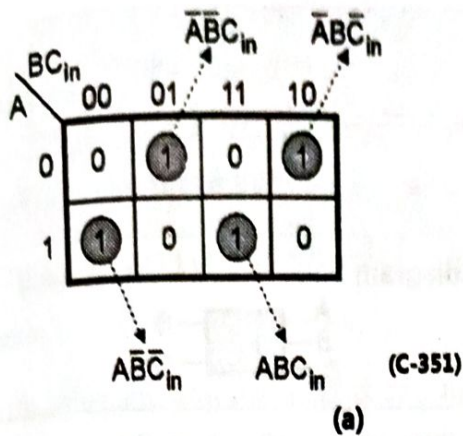
(b) Truth table

(C-350) Fig. 6.2.7 : Full adder

K-maps :

K-maps for the sum (S) and carry out (C_o) outputs and the corresponding Boolean expressions are as shown in Fig. 6.2.8.

K-map for the sum output



K-map for C_o output

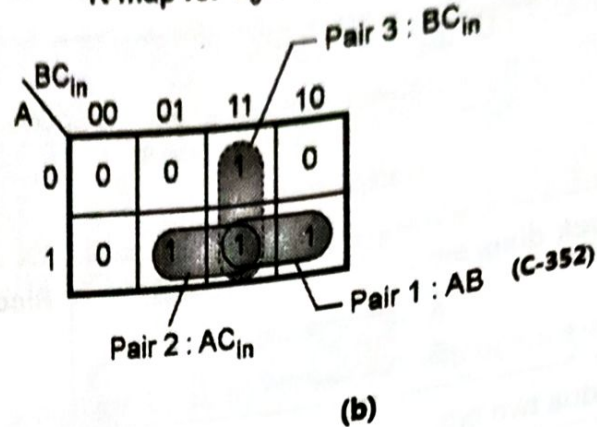


Fig. 6.2.8

Expression for sum output :

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}C_{in} + AB\bar{C}_{in}$$

$$S = C_{in}(\underbrace{\bar{A}\bar{B} + AB}_{\text{EX-NOR}}) + \bar{C}_{in}(\underbrace{\bar{A}B + A\bar{B}}_{\text{EX-OR}})$$

(C-6375)

$$\therefore S = C_{in}(\bar{A}B + A\bar{B}) + \bar{C}_{in}(\bar{A}\bar{B} + AB)$$

$$\text{Let } X = \bar{A}B + A\bar{B}$$

$$\therefore S = C_{in}\bar{X} + \bar{C}_{in}X = C_{in} \oplus X$$

$$\therefore S = C_{in} \oplus (\bar{A}B + A\bar{B})$$

$$\text{But } \bar{A}B + A\bar{B} = A \oplus B$$

$$\therefore S = C_{in} \oplus A \oplus B$$

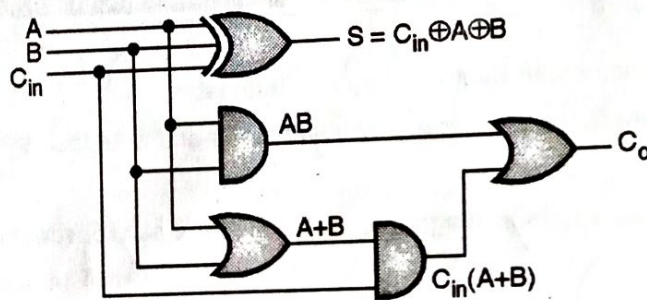
...(6.2.2)

Expression for carry output :

$$\text{For carry output } C_o = AB + A C_{in} + B C_{in} = AB + C_{in}(A + B)$$

...(6.2.3)

Logic diagram for full adder :



(C-3125) Fig. 6.2.8(c) : Full adder circuit

**6.2.3 Comparison of Half Adder and Full Adder :**

SPPU : Oct. 07, April 08

University Questions**Q. 1** Differentiate between half adder and full adder.

(Oct. 07, April 08)

Sr. No.	Half adder	Full adder
1.	Block diagram : (C-3467) 	Block diagram : (C-3468) 
2.	It adds two bits.	It adds three bits including carry C_{in} .
3.	It is used as a building block of a full adder.	It is used as a basic building block of an n-bit parallel adder.
4.	It has no practical applications.	Practically used in n-bit adders, ALUs etc.

6.3 Binary Subtractors :

The rules of binary subtraction are as follows :

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ with borrow } 1$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

Note that in the second case ($0 - 1$) it is necessary to borrow a 1.**6.3.1 Types of Binary Subtractors :**

- The types of binary subtractors are :

1. Half subtractor
2. Full subtractor

6.3.1.1 Half Subtractor :

SPPU : April 12, April 13, April 14, Oct. 14, Oct. 15, April 16, April 18

University Questions

- Q. 1** Explain half subtractor with neat diagram and truth table. (April 12, 4 Marks)
- Q. 2** Draw logic diagram for half adder and half subtractor and write their truth tables. (April 13, 4 Marks)
- Q. 3** Draw the truth table and logic diagram of half adder and half subtractor. (April 14, Oct. 15, April 18, 4 Marks)
- Q. 4** Draw the logic diagram of half subtractor. (Oct. 14, 2 Marks)
- Q. 5** Draw the circuit diagram of Half adder and Half subtractor and write truth tables of both. (April 16, 4 Marks)

Definition and Block Diagram :

Half subtractor is a combinational circuit with two inputs and two outputs (difference and borrow) as shown in its block diagram of Fig. 6.3.1.

It produces the difference between the two binary bits at the input and also produces an output (Borrow) to indicate if a 1 has been borrowed.

In the subtraction ($A - B$), A is called as **minuend bit** and B is called as **subtrahend bit**.

Truth table :

Truth table showing the outputs of a half subtractor for all the possible combinations of input are shown in Table 6.3.1.

Table 6.3.1 : Truth table for half subtractor

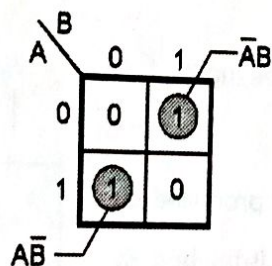
Inputs		Outputs	
A	B	Difference D ($A - B$)	Borrow B_0
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



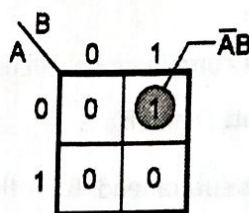
(C-3514) Fig. 6.3.1

K-maps for difference and borrow outputs :

The K-maps for the two outputs of a half subtractor are as shown in Fig. 6.3.2.



(a) K-map and simplification for difference output



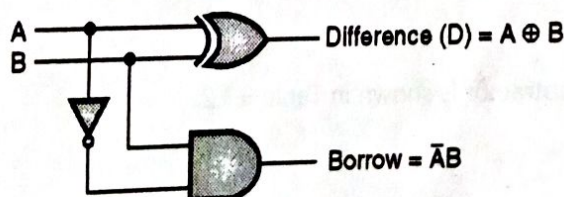
(b) K-map and simplification for the borrow output

(C-356) Fig. 6.3.2

$$\therefore \text{Difference } D = \bar{A}B + A\bar{B} = A \oplus B$$

$$\therefore \text{Borrow } B_0 = \bar{A}B$$

Logic diagram :



(C-357) Fig. 6.3.2(c) : Half subtractor circuit



7.1 Introduction :

- Now-a-days several combinational circuits are available in the integrated (MSI) form and are easily available in market.
- Depending on the level of complexity the integrated circuits are classified into four categories as follows :
 1. SSI : Small Scale Integration
 2. MSI : Medium Scale Integration
 3. LSI : Large Scale Integration
 4. VLSI : Very Large Scale Integration
- Due to availability of combinational circuits in the IC form; the design of combinational circuits has been simplified to a great extent. It also improves the reliability of the system because we have to use less number of external wired connections.
- In this chapter we are going to discuss the MSI devices such as multiplexer, demultiplexer, parity generators priority encoders, decoders etc.

7.2 Multiplexer (Data Selector) :

University Questions

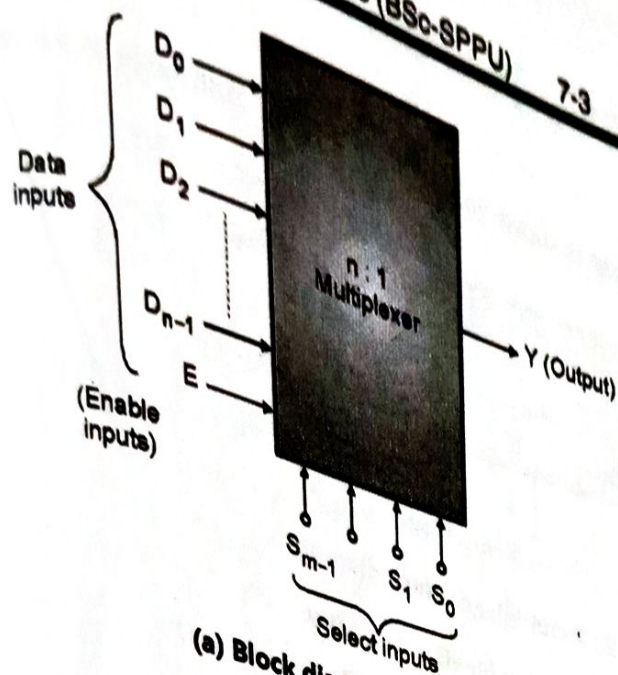
SPPU : April 13, Oct. 15

- Q. 1 What is the importance of strobe input in a multiplexer ? What does active low strobe mean ?
(April 13)
- Q. 2 What is multiplexer ? Draw circuit diagram of 8 : 1 multiplexer. Explain its working in brief.
(Oct. 15)

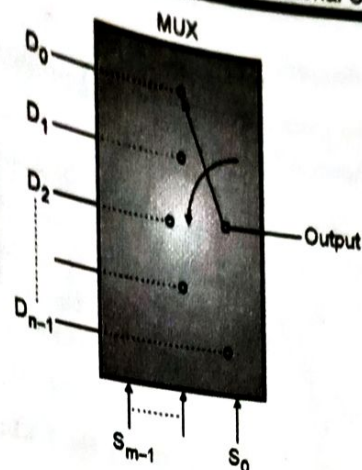
Definition and Block diagram :

- A multiplexer is a combinational circuit with n data inputs one output and m select inputs, which selects one of the n data inputs and routes (connects) it to the output. The selection of one of the n inputs is done with the help of the select inputs.
- Multiplexer is a special type of combinational circuit. The block diagram of an n -to-1 multiplexer is shown in Fig. 7.2.1(a) and its equivalent circuit is shown in Fig. 7.2.1(b).
- As shown, there are n -data inputs ($D_0, D_1, \dots, D_{n-1}$), one output Y and m select inputs, $S_0, S_1, \dots, S_{m-1}$. The relation between the number of data inputs (n) and number of select inputs (m) is as follows :

$$n = 2^m$$
- A multiplexer is a digital circuit which selects one of the n data inputs and routes (connects) it to the output. The selection of one of the n inputs is done with the help of the select inputs.
- To select n inputs we need m select lines such that $2^m = n$. Depending on the digital code applied at the select inputs, one out of n data sources is selected and transmitted to the single output Y .



(a) Block diagram of an $n:1$ multiplexer
(C-413) Fig. 7.2.1



(b) Equivalent circuit

- E is called as a **strobe or enable input** which is useful for cascading. It is generally an active low terminal, that means it will perform the required operation when it is low.
- As shown in Fig. 7.2.1(b) the multiplexer acts like a digitally controlled single pole, multiple way switch. The output gets connected to only one of the n data inputs at given instant of time.

7.2.1 Need of Multiplexers :

- In most of the electronic systems, the digital data is available from more than one sources. It is necessary to route this data over a single line.
- Under such circumstances we require a circuit which selects one of the many sources at a time.
- This circuit is nothing else but a multiplexer, which has many inputs, one output and some select inputs.
- Multiplexer improves the reliability of the digital system because it reduces the number of external wired connections.

7.3 Types of Multiplexers :

The types of multiplexers are :

1. $2:1$ multiplexer
2. $4:1$ multiplexer
3. $8:1$ multiplexer
4. $16:1$ multiplexer
5. $32:1$ multiplexer

7.3.1 $2:1$ Multiplexer :

SPPU : Oct. 14

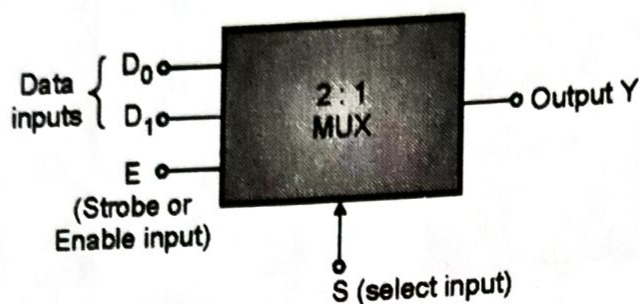
University Questions

Q. 1 Draw the logic diagram of 2 to 1 multiplexer.

(Oct. 14, 2 Marks)

Block diagram :

- The block schematic of a 2 : 1 multiplexer is shown in Fig. 7.3.1, and its truth table of this MUX is shown in Table 7.3.1.

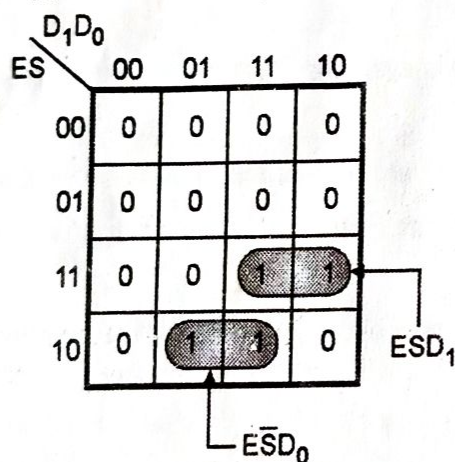


(C-414) Fig. 7.3.1 : 2 : 1 multiplexer block diagram

(C-414(a)) Table 7.3.1 : Truth table of a 2 : 1 MUX

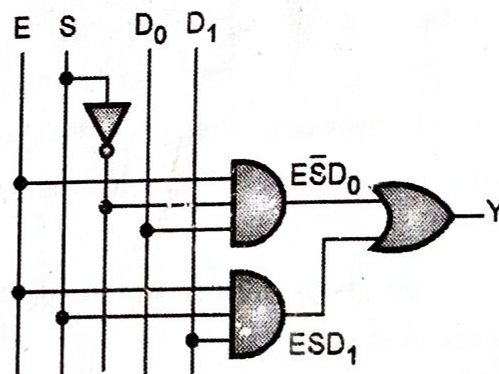
Enable E	Select S	D_1	D_0	Output Y
0	x	x	x	0
1	0	x	0	0
1	0	x	1	1
1	1	0	x	0
1	1	1	x	1

- Fig. 7.3.2(a) shows the k-map and Fig. 7.3.2(b) shows the realization.



$$\therefore Y = E\bar{S}D_0 + ESD_1$$

(a) k-map



(b) Realization

(C-3128) Fig. 7.3.2 : 2 : 1 MUX

7.3.2 4 : 1 Multiplexer :

SPPU : April 04, April 05, Oct. 10, April 15, April 18

University Questions

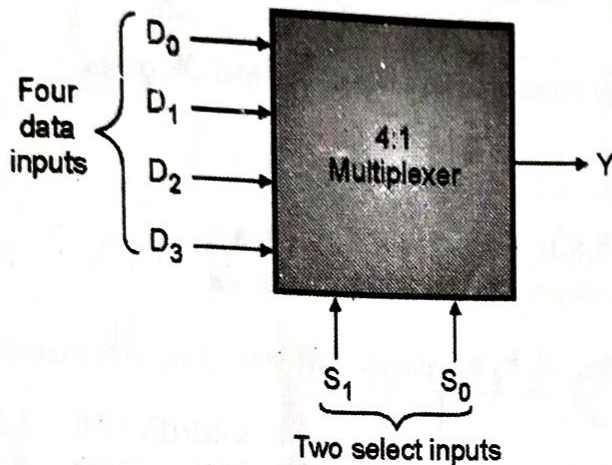
Q. 1 Explain function of a 4 : 1 multiplexer with logic diagram. (April 04, April 05, Oct. 10)

Q. 2 Draw and explain the working of 4 : 1 multiplexer built using AND-OR gates.

(April 15, April 18, 4 Marks)

Block diagram :

- Fig. 7.3.3 shows the block diagram of a 4 : 1 multiplexer and Table 7.3.2 gives its truth table..It has four data inputs, two select inputs and one output as shown.
- The strobe or enable terminal (G) has not been included to reduce the complexity.



(C-416)(a) Table 7.3.2 : Truth table

Select Inputs		Output Y
S ₁	S ₀	
0	0	D ₀
0	1	D ₁
1	0	D ₂
1	1	D ₃

(C-416) Fig. 7.3.3 : 4 : 1 multiplexer

- Note that $n = 4$ hence number of select-lines i.e. $m = 2$ so that $2^m = n$. The strobe terminal (G) has not been included to reduce the complexity.

Operation :

- The truth table tells us that if $S_1 S_0 = 00$, the data bit D_0 is selected and routed to output

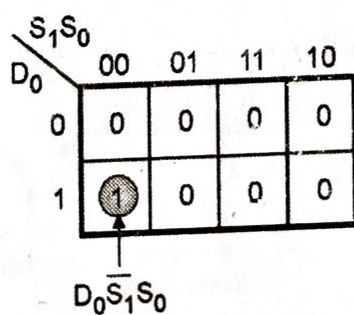
$$\therefore Y = D_0 \quad \dots \text{when } S_1 S_0 = 00$$

- Similarly if $S_1 S_0 = 01$, then D_1 is selected and routed to the output

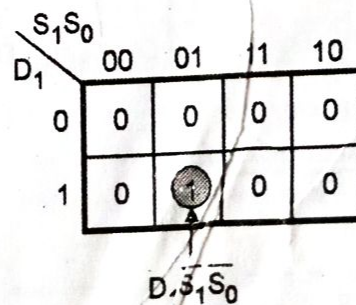
$$\therefore Y = D_1 \quad \dots \text{when } S_1 S_0 = 01$$

$$Y = D_2 \text{ for } S_1 S_0 = 10 \text{ and } Y = D_3 \text{ for } S_1 S_0 = 11$$

- The K-map is as shown in Fig. 7.3.4(a) for Y when $S_1 S_0 = 00$.



(a)



(b)

(C-3129) Fig. 7.3.4

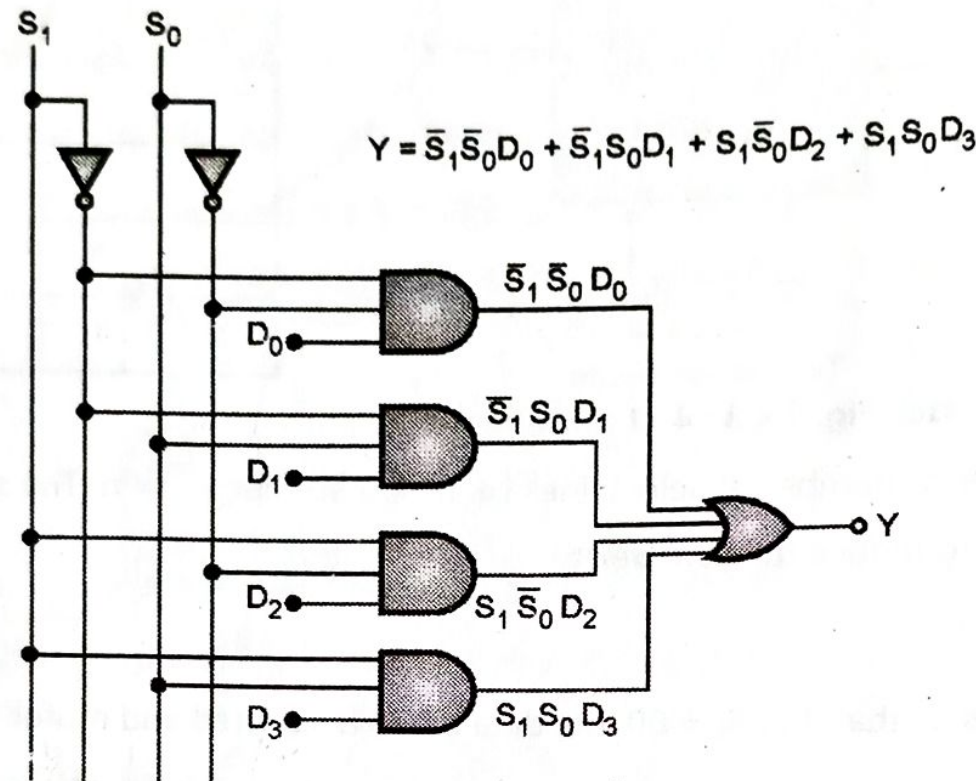


- Similarly we can draw k-map for $S_1 S_0 = 01$ as shown in Fig. 7.3.4(b). We can similarly draw the k-maps for $S_1 S_0 = 10$ and $S_1 S_0 = 11$. Fig. 7.3.5 shows the realization of 4 : 1 MUX using basic gates. Note that the expression for output Y is as follows :

$$Y = D_0 \bar{S}_1 \bar{S}_0 + D_1 \bar{S}_1 S_0 + D_2 S_1 \bar{S}_0 + D_3 S_1 S_0$$

Logic diagram :

Fig. 7.3.5 shows the logic diagram of a 4:1 Multiplexer using NOT, AND and OR gates



(C-417) Fig. 7.3.5 : Realization of 4 : 1 multiplexer using basic gates



7.7 Types of Demultiplexers :

Similar to the multiplexers, the demultiplexers are classified as follows :

1. 1 : 2 demultiplexer
2. 1 : 4 demultiplexer
3. 1 : 8 demultiplexer
4. 1 : 16 demultiplexer

SPPU : Oct. 12, April 14, April 16

7.7.1 1 : 2 Demultiplexer :

University Questions

- Q.1 Write truth table of 1 : 2 demultiplexer.
 Q.2 With logic diagram explain the working of 1 : 2 demultiplexer.
 Q.3 Describe the working of 1 : 2 Demultiplexer.

(Oct. 12, 2 Marks)

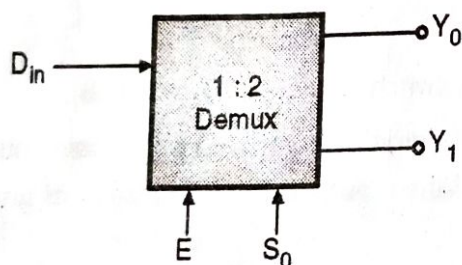
(April 14, 4 Marks)

(April 16, 4 Marks)

Block Diagram and Truth Table :

- The block diagram of 1 : 2 demultiplexer is shown in Fig. 7.7.1. It has one data input D_{in} , one select input S_0 , one enable (E) input and two outputs Y_0 and Y_1 .
- D_{in} is connected to Y_0 if $S_0 = 0$ and $E = 1$. Similarly D_{in} is connected to Y_1 if $S_0 = 1$ and $E = 1$.
- If $E = 0$, then both the outputs will be 0 irrespective of the inputs, because the DEMUX is disabled.
- The truth table of the 1 : 2 demultiplexer is as follows :

(c-7620) Table 7.7.1 : Truth table of demux 1 : 2



(c-448) Fig. 7.7.1

Enable	Select	Outputs	
		Y_1	Y_0
0	X	0	0
1	0	0	D_{in}
1	1	D_{in}	0

Realization of 1 : 2 demux :

Step 1 : To realize the 1 : 2 demultiplexer using gates we have to write the truth table as given in Table 7.7.2.

c-6188) Table 7.7.2 : Detail truth table of demux 1 : 2

Enable	Data	Select	Outputs	
E	D_{in}	S_0	Y_1	Y_0
0	X	X	0	0
1	0	0	0	0
1	1	0	0	1
1	0	1	0	0
1	1	1	1	0

$\left. \begin{array}{l} \text{Row 4: } Y_0 \text{ is connected to } D_{in} \\ \text{Row 5: } Y_1 \text{ is connected to } D_{in} \end{array} \right\}$

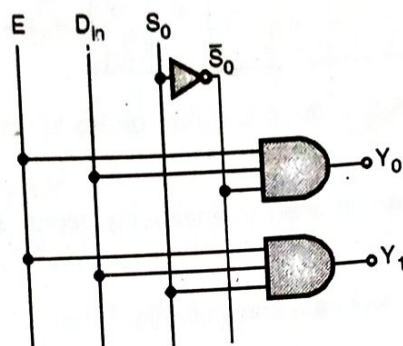
**Step 2 : Write the K maps :**

The K maps for the two outputs are as follows.

		Output Y_1						Output Y_0			
E	$D_{in} S_0$	00	01	11	10	E	$D_{in} S_0$	00	01	11	10
	0	0	0	0	0		0	0	0	0	0
	1	0	0	1	0		1	0	0	0	1

$Y_1 = E D_{in} S_0$ $Y_0 = E D_{in} \bar{S}_0$

(C-449) Fig. 7.7.1(a) : K maps



(C-450) Fig. 7.7.2 : 1 : 2 demux using gates

Thus the expressions for Y_0 and Y_1 are as follows :

$$Y_0 = E D_{in} \bar{S}_0 \quad \text{and} \quad Y_1 = E D_{in} S_0$$

The 1 : 2 demux using gates is shown in Fig. 7.7.2.

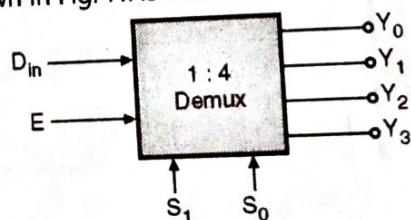
SPPU : April 13, Oct. 13, Oct. 14, Oct. 15

7.7.2 1 : 4 Demultiplexer :**University Questions**

- Q. 1 Draw logic diagram and explain with truth table 1 : 4 demultiplexer. How demultiplexer circuit can be converted to a decoder ? (April 13, 8 Marks)
- Q. 2 Draw logic diagram and explain the working of 1 : 4 demultiplexer. Write the truth table for the same. How demultiplexer is different than decoder ? (Oct. 13, 8 Marks)
- Q. 3 Explain working of a 1 : 4 demultiplexer with logic diagram. (Oct. 14, 4 Marks)
- Q. 4 Explain the working of 1 : 4 demultiplexer. What is the function of strobe input ? (Oct. 15, 4 Marks)

Block diagram and Truth Table :

The 1 : 4 demultiplexer is shown in Fig. 7.7.3 and its truth table is given in Table 7.7.3.



(C-451) Fig. 7.7.3 : Block diagram of 1 : 4 demultiplexer



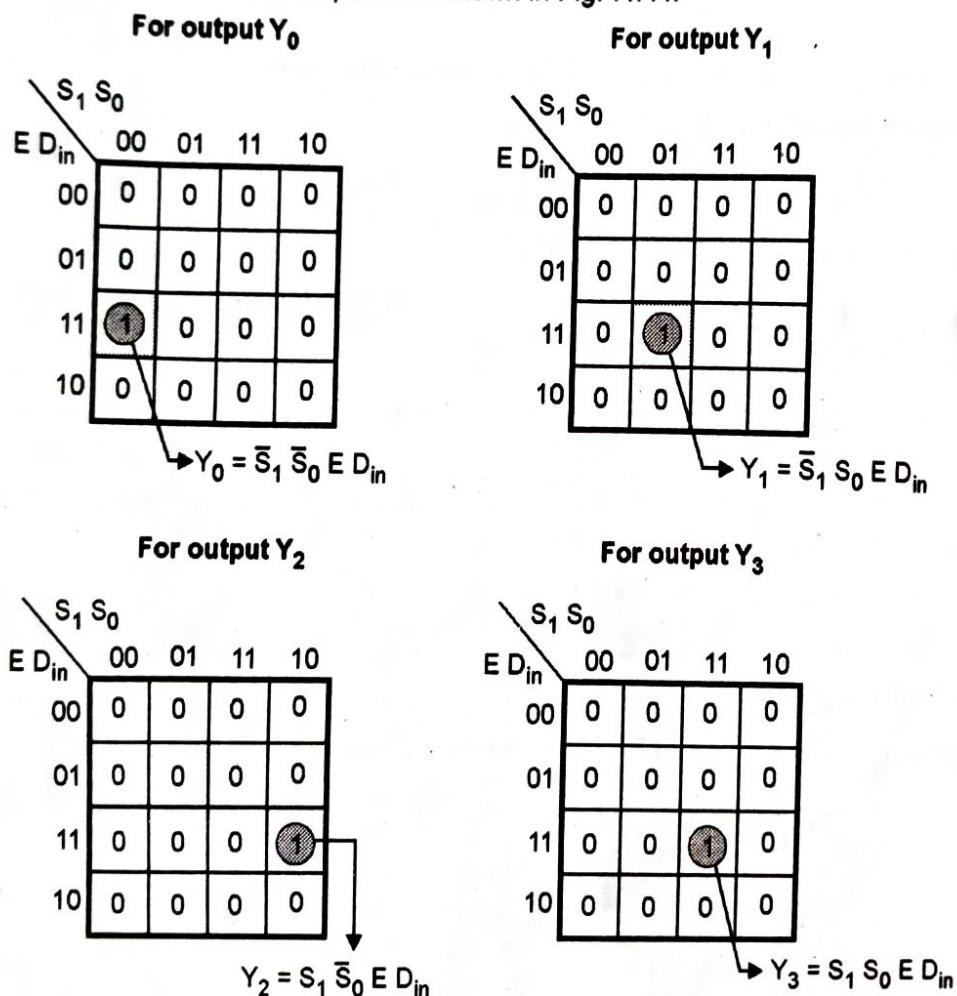
(C-6189) Table 7.7.3 : Truth table for 1 : 4 demultiplexer

		Inputs		Outputs			
E	D _{in}	S ₁	S ₀	Y ₀	Y ₁	Y ₂	Y ₃
1	0	0	0	0	0	0	0
1	1	0	0	1	0	0	0
1	0	0	1	0	0	0	0
1	1	0	1	0	1	0	0
1	0	1	0	0	0	0	0
1	1	1	0	0	0	1	0
1	0	1	1	0	0	0	0
1	1	1	1	0	0	0	1

Y_0 is connected to D_{in}
 Y_1 is connected to D_{in}
 Y_2 is connected to D_{in}
 Y_3 is connected to D_{in}

- D_{in} is connected to Y_0 when $S_1 S_0 = 00$, it is connected to Y_1 when $S_1 S_0 = 01$ and so on. The other outputs will remain zero.
- The enable input needs to be high in order to enable the demux. If $E = 0$ then all the outputs will be low irrespective of everything.

K-maps : The K-maps for the four outputs are shown in Fig. 7.7.4.

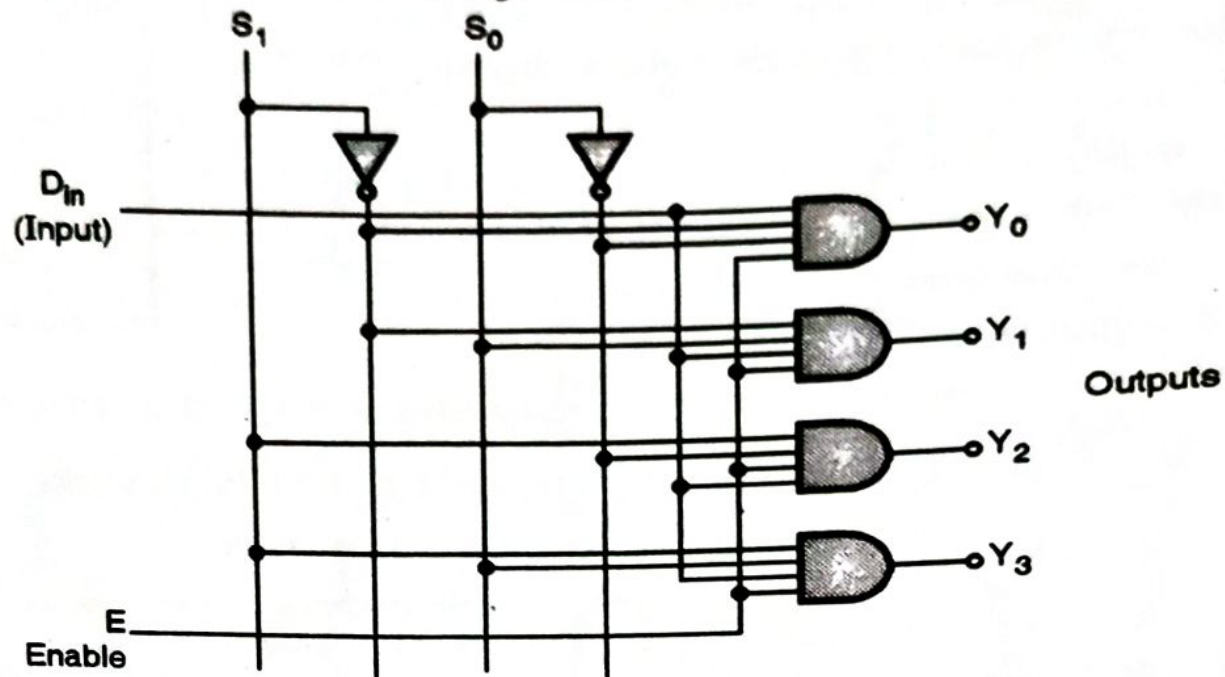


(C-452) Fig. 7.7.4 : K-maps for various outputs



Implementation :

The 1 : 4 demux is implemented as shown in Fig. 7.7.5.



(C-453) Fig. 7.7.5 : 1 : 4 demultiplexer

7.8 Comparison of Multiplexer and Demultiplexer :

Table 7.8.1 : Comparison of MUX and DEMUX

Sr. No.	Parameter	Multiplexer	Demultiplexer
1.	Type of logic circuit	Combinational	Combinational
2.	Number of data inputs	n	1
3.	Number of select inputs	m	m
4.	Number of data output	1	n
5.	Relation between input / output lines and select lines	$n = 2^m$	$n = 2^m$
6.	Operation principle	Many to 1 or as data selector	1 to many or data distributor

Sr. No.	Parameter	Multiplexer	Demultiplexer
7.	Application	- As a universal logic circuit we can implement any combinational circuit using a MUX. - In time division multiplexing at the sending end.	- We can implement some combinational circuit. - In TDM system at receiving end.

7.8.1 Applications of Demultiplexer :

University Questions

SPPU : April 09, April 16

Q. 1 Write in brief about 2 applications of multiplexer and demultiplexer each.

(April 09, April 16, 2 Marks)

1. As a decoder.
2. As a one to many switch.
3. For implementation of Boolean expressions.

7.9 Encoders :

SPPU : Oct. 06, April 08, April 09, Oct. 11, Oct. 12, Oct. 13

University Questions

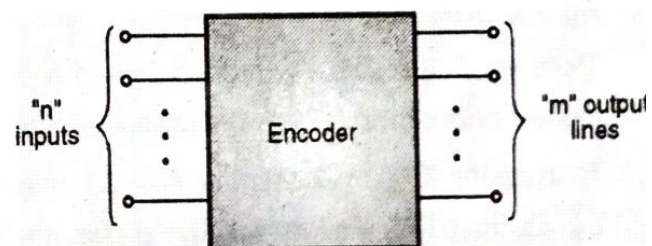
- Q. 1** What is encoding ? (Oct. 06, April 08, April 09, Oct. 11)
- Q. 2** What is encoding ? With logic diagram explain the working of decimal to binary encoder. (Oct. 06)
- Q. 3** Define encoder. With logic diagram explain octal to binary encoder. What is priority encoder ? (Oct. 12, 8 Marks)
- Q. 4** What is an encoder ? What is the function of strobe input of encoder ? (Oct. 13, 2 Marks)

Definition :

Encoder is a combinational circuit which is designed to perform the inverse operation of the decoder.

Block Diagram :

- Block diagram of an encoder is shown in Fig. 7.9.1.
- An encoder has "n" number of input lines and "m" number of output lines.



(C-466) Fig. 7.9.1 : Block diagram of an encoder